Product-Company Comeback Plan

Six weeks • 26 October–6 December 2026 • DSA + Java/Spring • Claude's original assignment compared with a more focused recommendation

How to use this plan: Claude's columns preserve the original plan's direction. The recommendation columns are adjusted for a learner who understands Java while reading but struggles to code independently. Prioritise writing code from a blank file, testing edge cases and revisiting weak problems. Advanced items marked optional are not failures if deferred.

Week 1 — Setup, Big-O and Hash Maps

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 26 Oct Day 1	Setup; learn Big-O; LC 217 Contains Duplicate: compare brute force, sorting and HashSet.	Complete LC 217 from a blank file. Explain O(n^2), O(n log n) and O(n) approaches. Set up a tracker.	Write operation-complexity cheat sheet for ArrayList, LinkedList, HashMap, HashSet, TreeMap, ArrayDeque and PriorityQueue.	Implement simple List and Set examples. Handwrite a small Big-O table; digitally save runnable examples.	[ ] DSA [ ] Java [ ] Review
Tue, 27 Oct Day 2	LC 242 Valid Anagram; revisit 217 from scratch.	Solve LC 242 using a frequency array first, then HashMap if time allows. Re-code 217 without notes.	HashMap/HashSet API drill: getOrDefault, merge, computeIfAbsent, putIfAbsent and entrySet; 10 snippets.	Write 3 small HashMap exercises, including word frequency. Practise getOrDefault and merge.	[ ] DSA [ ] Java [ ] Review
Wed, 28 Oct Day 3	LC 1 Two Sum using one-pass HashMap; revisit 242.	Solve LC 1 from scratch. Test duplicates, negative values and no solution. Revisit 242 briefly.	equals/hashCode contract; use a custom class as a HashMap key and demonstrate mutable-key problem.	Implement a small Employee key class; explain why equals and hashCode must agree.	[ ] DSA [ ] Java [ ] Review
Thu, 29 Oct Day 4	LC 49 Group Anagrams using sorted-string key and character-count key; revisit 1.	Try sorted-string solution first. Treat the character-count key as a stretch goal.	HashMap internals: buckets, hash spreading, index calculation, load factor 0.75 and resize.	Understand buckets, collisions and resizing conceptually. Do not memorise implementation details.	[ ] DSA [ ] Java [ ] Review
Fri, 30 Oct Day 5	LC 347 Top K Frequent Elements using HashMap + min-heap, then bucket sort; revisit 49.	If LC 347 is too difficult, study the frequency map and PriorityQueue approach with a worked example.	Treeification thresholds, resize splitting, null key, thread safety; HashMap vs Hashtable vs ConcurrentHashMap.	Compare HashMap, TreeMap and ConcurrentHashMap in a short digital reference. Explain use cases.	[ ] DSA [ ] Java [ ] Review
Sat, 31 Oct Day 6	LC 238 Product of Array Except Self; timed revisits of 217, 1 and 49.	Attempt LC 238 after reviewing prefix/suffix products. Revisit only the weakest earlier problem.	String immutability, string pool, StringBuilder vs StringBuffer, == vs equals and intern. Career: five work stories.	Code StringBuilder examples and explain String comparison. Draft five truthful work-story bullets.	[ ] DSA [ ] Java [ ] Review
Sun, 01 Nov Day 7	LC 36 Valid Sudoku; revisit 347 and 238. Java Q&A Set 1; list 10 referral contacts; weekly review.	LC 36 is optional. Prefer re-coding two weak problems and explaining HashMap/HashSet.	Answer collections and strings questions aloud; complete weekly retrospective.	Answer 5 core Java questions aloud. Review notes and choose two weak topics for next week.	[ ] DSA [ ] Java [ ] Review

Weekly review: Which problems did I solve without hints? Which two patterns are weakest? Which Java topic can I explain but not implement? What will I revise next week?

Week 2 — Java Language Features and Two Pointers

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 02 Nov Day 8	LC 271 Encode and Decode Strings using length-prefix format; revisit 36.	Try LC 271; if stuck, practise parsing a length-prefixed string with a small example.	Lambdas, functional interfaces (Function, Predicate, Consumer, Supplier, BiFunction) and method references.	Write tiny examples of Predicate, Function and Consumer. Explain each in your own words.	[ ] DSA [ ] Java [ ] Review
Tue, 03 Nov Day 9	LC 128 Longest Consecutive Sequence; revisit 271.	Solve LC 128 using a HashSet; explain why counting starts only at sequence starts.	Streams I: map, filter, reduce, collect; Collectors toList, toMap, groupingBy and counting.	Use streams to filter a list, transform values and group words by length.	[ ] DSA [ ] Java [ ] Review
Wed, 04 Nov Day 10	LC 125 Valid Palindrome with two pointers; revisit 128.	Solve LC 125 independently. Practise pointer movement and character checks.	Streams II: flatMap, Comparator.comparing/thenComparing, Optional and parallel-stream pitfalls.	Sort Employee objects by name and salary. Use Optional in a simple example.	[ ] DSA [ ] Java [ ] Review
Thu, 05 Nov Day 11	LC 167 Two Sum II on sorted input; revisit 125.	Solve LC 167 and explain why the sorted input permits two pointers.	Records, sealed classes, pattern matching, text blocks and var.	Learn records and var with small runnable snippets. Treat sealed classes as overview only.	[ ] DSA [ ] Java [ ] Review
Fri, 06 Nov Day 12	LC 15 3Sum; revisit 167 and 1.	Attempt LC 15; focus on sorting, two pointers and duplicate skipping. Revisit LC 1.	Checked vs unchecked exceptions, try-with-resources, custom exceptions and Java 21 virtual threads overview.	Write one custom exception and one try-with-resources example. Virtual threads: read-only overview.	[ ] DSA [ ] Java [ ] Review
Sat, 07 Nov Day 13	LC 11 Container With Most Water; timed revisits of 238, 347 and 36.	Solve LC 11; explain why the shorter side moves. Revisit one easy hashing problem.	Generics, wildcards, type erasure, Comparable vs Comparator. Rewrite resume.	Code a generic method and Comparator. Draft a one-page resume without invented metrics.	[ ] DSA [ ] Java [ ] Review
Sun, 08 Nov Day 14	LC 42 Trapping Rain Water (hard); revisit 15 and 11. Java Q&A Set 2.	Treat LC 42 as optional/stretch. If the approach is unclear after 30 minutes, study one solution and re-code later.	Answer Java language-feature questions and perform weekly review.	Explain lambdas, streams, exceptions and generics aloud. Review resume draft.	[ ] DSA [ ] Java [ ] Review

Weekly review: Which problems did I solve without hints? Which two patterns are weakest? Which Java topic can I explain but not implement? What will I revise next week?

Week 3 — Sliding Window, Stack and Concurrency Basics

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 09 Nov Day 15	LC 121 Best Time to Buy and Sell Stock; re-code LC 42 two-pointer approach.	Solve LC 121 using minimum-so-far. Revisit LC 42 only if earlier basics are stable.	Thread lifecycle, Runnable vs Callable, join and daemon threads.	Understand process vs thread, Runnable and Callable. No advanced threading implementation yet.	[ ] DSA [ ] Java [ ] Review
Tue, 10 Nov Day 16	LC 3 Longest Substring Without Repeating Characters; revisit 121.	Solve LC 3 using a sliding window. Trace left/right pointers on a repeated-character example.	synchronized and intrinsic locks; write a racy two-thread counter and fix it.	Write a small synchronized-counter demo only if comfortable; otherwise explain race conditions.	[ ] DSA [ ] Java [ ] Review
Wed, 11 Nov Day 17	LC 424 Longest Repeating Character Replacement; revisit 3.	Practise LC 3 again before attempting LC 424. Learn the window-validity condition.	volatile, happens-before, Java Memory Model basics, AtomicInteger and CAS.	Understand volatile vs synchronized conceptually; write a short comparison in digital notes.	[ ] DSA [ ] Java [ ] Review
Thu, 12 Nov Day 18	LC 567 Permutation in String using fixed-size window and counts; revisit 424.	Attempt LC 567 using two frequency arrays. Revisit LC 424 if time permits.	ReentrantLock, ReadWriteLock and Condition; create and diagnose deadlock with jstack.	Learn the purpose of locks at a high level. Skip deliberate deadlock tooling if it consumes too much time.	[ ] DSA [ ] Java [ ] Review
Fri, 13 Nov Day 19	LC 76 Minimum Window Substring (hard); revisit 567.	LC 76 is optional. Spend the core session re-coding LC 3 or LC 567 and explaining window invariants.	wait/notify and BlockingQueue producer-consumer.	Understand producer-consumer and why BlockingQueue is useful. No need to master wait/notify yet.	[ ] DSA [ ] Java [ ] Review
Sat, 14 Nov Day 20	LC 239 Sliding Window Maximum and LC 20 Valid Parentheses; timed revisits.	Prioritise LC 20. Attempt LC 239 only if the deque pattern is manageable.	Implement producer-consumer with wait/notify and BlockingQueue. Update LinkedIn and DSA README.	Implement stack-based parentheses validation. Update LinkedIn and write a concise DSA README.	[ ] DSA [ ] Java [ ] Review
Sun, 15 Nov Day 21	LC 155 Min Stack; revisit 76 and 239. Java Q&A Set 3; weekly review.	Complete LC 155 if ready; otherwise re-code LC 20 and review stack operations.	Answer concurrency-I questions aloud.	Explain race condition, synchronized, volatile and BlockingQueue in plain language.	[ ] DSA [ ] Java [ ] Review

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Week 4 — Stack, Binary Search and Concurrency II

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 16 Nov Day 22	LC 150 Evaluate Reverse Polish Notation; revisit 20 and 155.	Solve LC 150 with a stack. Carefully handle operand order and integer division.	ExecutorService, thread pools and ThreadPoolExecutor parameters.	Understand ExecutorService and why thread pools exist. Defer deep parameter tuning.	[ ] DSA [ ] Java [ ] Review
Tue, 17 Nov Day 23	LC 739 Daily Temperatures using monotonic stack; revisit 150.	Solve LC 739 and explain what the stack indices represent.	Future vs CompletableFuture: supplyAsync, thenApply, thenCompose, thenCombine, allOf and exceptionally.	Learn Future vs CompletableFuture with one simple example; avoid memorising every method.	[ ] DSA [ ] Java [ ] Review
Wed, 18 Nov Day 24	LC 853 Car Fleet; revisit 739.	LC 853 is optional if the stack pattern is still new; re-code LC 739 first.	ConcurrentHashMap, CopyOnWriteArrayList and ConcurrentLinkedQueue.	Know when concurrent collections are useful. Focus on HashMap vs ConcurrentHashMap.	[ ] DSA [ ] Java [ ] Review
Thu, 19 Nov Day 25	LC 22 Generate Parentheses; revisit 853.	Try LC 22 as an introduction to backtracking; otherwise consolidate stack problems.	CountDownLatch, CyclicBarrier, Semaphore and ThreadLocal.	Write one-line use cases for the four utilities. Defer detailed implementations.	[ ] DSA [ ] Java [ ] Review
Fri, 20 Nov Day 26	LC 84 Largest Rectangle in Histogram (hard); revisit 22.	LC 84 is optional. Practise LC 150 and 739 under a short timer instead.	Virtual vs platform threads; blocking I/O and Spring Boot virtual-thread setting.	Understand virtual threads at a high level and when they may help.	[ ] DSA [ ] Java [ ] Review
Sat, 21 Nov Day 27	LC 704 Binary Search and LC 74 Search a 2D Matrix; timed stack revisits.	Prioritise LC 704 template and LC 74 only if time allows. Revisit LC 739.	Build a thread-safe bounded blocking queue and token-bucket rate limiter; send five connection messages.	Skip building advanced concurrency utilities. Implement binary search in Java and send five networking messages.	[ ] DSA [ ] Java [ ] Review
Sun, 22 Nov Day 28	LC 875 Koko Eating Bananas; revisit 704 and 74. Java Q&A Set 4.	Solve LC 875 only if binary search fundamentals are clear; otherwise re-code LC 704.	Answer concurrency-II questions; weekly review.	Explain thread pools, Future, ConcurrentHashMap and virtual threads at interview level.	[ ] DSA [ ] Java [ ] Review

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Week 5 — Binary Search, Linked Lists and JVM Basics

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 23 Nov Day 29	LC 153 Find Minimum in Rotated Sorted Array; revisit 875.	Solve LC 153 and practise identifying the sorted half.	JVM architecture: class loading, heap, thread stacks, metaspace and program counter.	Draw a simple JVM memory diagram by hand; explain heap vs stack.	[ ] DSA [ ] Java [ ] Review
Tue, 24 Nov Day 30	LC 33 Search in Rotated Sorted Array; revisit 153.	Attempt LC 33; re-code LC 153 if the rotated-array logic is unclear.	Heap generations: Eden, survivor, old; minor vs major GC and object promotion.	Understand garbage collection and object reachability conceptually.	[ ] DSA [ ] Java [ ] Review
Wed, 25 Nov Day 31	LC 981 Time Based Key-Value Store; revisit 33.	LC 981 is optional/stretch. Review binary search and write a simple timestamp lookup if needed.	G1, ZGC and Parallel GC; stop-the-world pauses; -Xms, -Xmx and key JVM flags.	Learn what heap-size flags mean. Do not memorise collector internals.	[ ] DSA [ ] Java [ ] Review
Thu, 26 Nov Day 32	LC 206 Reverse Linked List; read-only stretch: LC 4 Median of Two Sorted Arrays; revisit 981.	Prioritise LC 206 with iterative pointer changes. Skip LC 4.	Memory leaks: static collections, unclosed resources, ThreadLocal, listeners and OutOfMemoryError types.	Draw linked-list pointer changes. Understand common memory-retention causes.	[ ] DSA [ ] Java [ ] Review
Fri, 27 Nov Day 33	LC 21 Merge Two Sorted Lists; revisit 206.	Solve LC 21 using a dummy node; re-code LC 206 without notes.	JIT, tiered compilation and escape analysis at awareness level.	Know that JIT compiles frequently executed bytecode; do not dive into compiler internals.	[ ] DSA [ ] Java [ ] Review
Sat, 28 Nov Day 34	LC 141 Linked List Cycle and LC 19 Remove Nth Node; timed revisits.	Prioritise LC 141 and LC 19. Attempt both only after revising pointer basics.	Run a Spring Boot app with GC logging, take heap dump and inspect in VisualVM/MAT. Research 10 job descriptions and message five contacts.	Build a small linked-list example. If tooling is unfamiliar, read about GC logs rather than spending hours setting up MAT.	[ ] DSA [ ] Java [ ] Review
Sun, 29 Nov Day 35	LC 143 Reorder List; revisit 141 and 19. Java Q&A Set 5; weekly review.	Try LC 143 only if the three pointer steps are clear; otherwise re-code LC 206 and 21.	Answer JVM questions aloud.	Explain heap, stack, GC and memory leaks in simple terms.	[ ] DSA [ ] Java [ ] Review

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Week 6 — Spring Internals, Security and Checkpoint

Date	DSA — Claude original	DSA — My recommendation	Java — Claude original	Java — My recommendation	Done
Mon, 30 Nov Day 36	LC 2 Add Two Numbers; revisit 143.	Attempt LC 2 if linked-list fundamentals are solid; otherwise revisit LC 206/21.	Spring IoC/DI, bean scopes and lifecycle, @PostConstruct, BeanPostProcessor, constructor vs field injection, circular dependencies.	Code a tiny Spring service using constructor injection; explain IoC and DI.	[ ] DSA [ ] Java [ ] Review
Tue, 01 Dec Day 37	LC 287 Find the Duplicate Number using Floyd's cycle detection or binary search; revisit 2.	LC 287 is optional/stretch. Re-code a familiar linked-list problem if pointer logic remains shaky.	Spring Boot auto-configuration, @SpringBootApplication, @Conditional annotations, starters and profiles.	Create or inspect a small Spring Boot app; explain profiles and auto-configuration.	[ ] DSA [ ] Java [ ] Review
Wed, 02 Dec Day 38	LC 138 Copy List with Random Pointer; revisit 287.	Treat LC 138 as optional. Prioritise one previously solved medium problem from scratch.	AOP/proxies; @Transactional propagation, isolation, rollback rules and self-invocation pitfall.	Understand proxies and why self-invocation can bypass @Transactional. Write a short example.	[ ] DSA [ ] Java [ ] Review
Thu, 03 Dec Day 39	LC 146 LRU Cache using HashMap + doubly linked list, then LinkedHashMap; revisit 138.	LC 146 is optional/stretch. Focus on explaining HashMap and linked-list components.	Spring Security filter chain, JWT validation, OAuth2/OIDC with Azure AD; rehearse production auth issue.	Draw request-to-authentication flow and explain the real production issue without confidential details.	[ ] DSA [ ] Java [ ] Review
Fri, 04 Dec Day 40	LC 23 Merge k Sorted Lists using PriorityQueue or divide-and-conquer; revisit 146.	LC 23 is optional. Re-code LC 21 or LC 141 and explain complexity.	JPA/Hibernate N+1, lazy vs eager, first/second-level cache, dirty checking and optimistic/pessimistic locking.	Build a tiny JPA example or explain N+1, transactions and lazy loading with a practical scenario.	[ ] DSA [ ] Java [ ] Review
Sat, 05 Dec Day 41	LC 25 Reverse Nodes in k-Group (hard); timed revisits of 23, 21 and 19. Mock interview.	Skip LC 25 if fundamentals need work. Do one familiar medium under timed conditions and review mistakes.	Prepare 2-minute and 5-minute project pitches. Mock interview: one medium problem plus five Java questions.	Practise a truthful project pitch: architecture, your role, one difficult bug and verified impact.	[ ] DSA [ ] Java [ ] Review
Sun, 06 Dec Day 42	Checkpoint: choose two problems from LC 209, 496, 34, 75, 160, 234, 680 and 692; Java Q&A and retrospective.	Test two familiar problems in 60 minutes. At least one should be solved without hints; record gaps honestly.	60-minute coding test, 20-minute Java Q&A, 30-minute retrospective. Decide whether to proceed or consolidate.	Answer ten Java/Spring questions aloud. Repeat weak topics next week rather than rushing into advanced DSA.	[ ] DSA [ ] Java [ ] Review

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